University of Illinois Urbana Champaign

Applied Nonlinear Programming

IE 510

Final Project

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Task 1

Choose ℓ, i, j, define

$$B_\ell^{ij}(y) = \max\left\{ y_{\ell i} - v_{\ell i},\ y_{\ell j} + v_{\ell i},\ \max_{r\in[n]\setminus\{i,j\}} y_{\ell r} \right\}.$$

Then the convex part can be written as

$$g(y) = \max_{\substack{i\neq j\in[n]\\ k\in[m]}} \left(y_{kj} + \sum_{\ell\in[m]\setminus\{k\}} B_\ell^{ij}(y) \right).$$

We want to show

$$\min_{y\in\mathbb{R}^{mn}} f(y)$$

is equivalent to

$$\begin{aligned}
\min_{y,z,w}\quad & w - h(y) \\
\text{s.t.}\quad & y_{kj} + \sum_{\ell\in[m]\setminus\{k\}} z_{\ell ij} - w \le 0, && \forall k\in[m],\ i\neq j\in[n], \\
& y_{\ell i} - z_{\ell ij} \le v_{\ell i}, && \forall \ell\in[m],\ i\neq j\in[n], \\
& y_{\ell j} - z_{\ell ij} \le -v_{\ell i}, && \forall \ell\in[m],\ i\neq j\in[n], \\
& y_{\ell r} - z_{\ell ij} \le 0, && \forall \ell\in[m],\ i\neq j\in[n],\ r\in[n]\setminus\{i,j\}.
\end{aligned}$$

Notice

$$y_{\ell i} - z_{\ell ij} \le v_{\ell i} \equiv z_{\ell ij} \ge y_{\ell i} - v_{\ell i}.$$

And,

$$y_{\ell j} - z_{\ell ij} \le -v_{\ell i} \equiv z_{\ell ij} \ge y_{\ell j} + v_{\ell i}.$$

And,

$$y_{\ell r} - z_{\ell ij} \le 0 \equiv z_{\ell ij} \ge y_{\ell r}, \qquad r\in[n]\setminus\{i,j\}.$$

$$\therefore z_{\ell ij} \ge \max\left\{ y_{\ell i} - v_{\ell i},\ y_{\ell j} + v_{\ell i},\ \max_{r\in[n]\setminus\{i,j\}} y_{\ell r} \right\} = B_\ell^{ij}(y).$$

The remaining constraint

$$y_{kj} + \sum_{\ell\in[m]\setminus\{k\}} z_{\ell ij} - w \le 0 \equiv w \ge y_{kj} + \sum_{\ell\in[m]\setminus\{k\}} z_{\ell ij}.$$

Meaning, $\forall\ y$ fixed, the smallest feasible value of $z_{\ell ij}$ is

$$z_{\ell ij} = B_\ell^{ij}(y),$$

and it follows from that that

$$w = \max_{\substack{i \neq j \in [n] \\ k \in [m]}} \left(y_{kj} + \sum_{\ell \in [m] \setminus \{k\}} B_\ell^{ij}(y) \right) = g(y).$$

$\therefore$ at the feasible choices of z and w for fixed y, the constrained objective becomes

$$w - h(y) = g(y) - h(y) = f(y).$$

Which shows minimizing $w - h(y)$ over the feasible set is equivalent to minimizing $f(y)$ over y. Also note that, w is linear (affine) and $h(y)$ is convex, so

$$w - h(y)$$

is concave. All the constraints are linear inequalities. So the new problem has a concave objective function and linear constraints. □

Task 2

An iterative algorithm is designed based on the constrained reformulation from Task 1. The objective $w - h(y)$ is concave and the feasible set is a polyhedron, so the global minimum is attained at an extreme point.

Successive linearization via the subgradient inequality

Since $h(y) = \sum_\ell \max_r y_{\ell r}$ is convex but non-differentiable, use the subgradient inequality: for any $y^k \in \mathbb{R}^{mn}$ and any $s^k \in \partial h(y^k)$,

$$h(y) \;\geq\; h(y^k) + \langle s^k,\, y - y^k \rangle, \qquad \forall y \in \mathbb{R}^{mn}. \tag{1}$$

A subgradient of h at y^k is given by an argmax in each row: pick

$$r_\ell^k \in \arg\max_{r \in [n]} y_{\ell r}^k \qquad \forall\, \ell \in [m],$$

and define s^k by $s_{\ell,r}^k = 1$ when $r = r_\ell^k$ and 0 otherwise. Then

$$\langle s^k, y \rangle = \sum_{\ell=1}^{m} y_{\ell,\, r_\ell^k} \in \partial h(y^k).$$

Negating (1) gives

$$-h(y) \;\leq\; -h(y^k) - \langle s^k, y - y^k \rangle,$$

an affine function larger than $-h$ that attains equality at $y = y^k$. Substituting this for $-h$ in the constrained problem from Task 1 leaves the linear program

$$\begin{aligned}
\min_{y,\,w,\,z} \quad & w \;-\; \sum_{\ell=1}^{m} y_{\ell,\,r_\ell^k} \\
\text{s.t.} \quad & y_{kj} + \sum_{\ell \in [m]\setminus\{k\}} z_{\ell ij} - w \le 0, && \forall k,\ i \neq j, \\
& y_{\ell i} - z_{\ell ij} \le v_{\ell i}, && \forall \ell,\ i \neq j, \\
& y_{\ell j} - z_{\ell ij} \le -v_{\ell i}, && \forall \ell,\ i \neq j, \\
& y_{\ell r} - z_{\ell ij} \le 0, && \forall \ell,\ i \neq j,\ r \notin \{i, j\}, \\
& y_{\ell,1} = 0, && \forall \ell \in [m].
\end{aligned} \tag{LP$_k$}$$

The next iterate y^{k+1} is the y-component of any optimal solution to (LP$_k$).

Descent property

We need $f(y^{k+1}) \le f(y^k)$. By Task 1, $f(y) = \min_{w,z}\{w - h(y) : (y, w, z) \text{ feasible}\}$, so

$$\begin{aligned}
f(y^{k+1}) \;&\le\; w^{k+1} - h(y^{k+1}) \\
&\le\; w^{k+1} - h(y^k) - \langle s^k,\, y^{k+1} - y^k \rangle && \text{(subgradient inequality (1))} \\
&=\; \left[w^{k+1} - \langle s^k, y^{k+1} \rangle\right] - h(y^k) + \langle s^k, y^k \rangle \\
&\le\; \left[w^k - \langle s^k, y^k \rangle\right] - h(y^k) + \langle s^k, y^k \rangle && \left(y^{k+1} \text{ minimizes (LP}_k\text{)}\right) \\
&=\; w^k - h(y^k) \;=\; f(y^k),
\end{aligned}$$

where we use $w^k = g(y^k)$ at the optimum of the LP from iteration $k-1$.

In the experiments the iteration above terminates in 2–3 LPs from any starting point, because the argmax pattern $r^k = (r_1^k, \ldots, r_m^k)$ stabilizes quickly: once $r^{k+1} = r^k$ no further descent is possible.

Therefore wrap (LP$_k$) in a two-phase global search.

- **Phase A.** Generate S random initializations $y^{(0,s)} \sim \mathcal{N}(0, I)$ with $y_{\ell,1}^{(0,s)} = 0$ in each row, run the LP iteration to convergence from each, and keep the best solution.
- **Phase B .** Take the argmax pattern $r^\star$ of the best iterate. Randomly flip $\kappa \in \{1, 2, 3\}$ rows of $r^\star$ to a different column to get $\tilde{r}$, use $\tilde{r}$ as the subgradient in a new LP , and run the iteration from the LP solution. Repeat T times, accepting any improvement.

Each Phase B is a sequence of (LP$_k$) solves and inherits the descent property, so the overall iterate sequence is monotonically non-increasing. The total number of LPs solved is the iteration count.

Algorithm

Algorithm 1:

Input: Cost matrix $v \in \mathbb{R}_+^{m \times n}$; S , T

Output: $\tilde{y}$, iteration count N

$f^\star \leftarrow +\infty$,;

$y^\star \leftarrow 0$,;

$N \leftarrow 0$;

for $s = 1, \ldots, S$ **do** `// Phase A`

  Sample $y^{(0)} \sim \mathcal{N}(0, I)$,;

  set $y^{(0)}_{\ell,1} \leftarrow 0$;

  while *not converged* **do**

    $r^k_\ell \leftarrow \arg\max_r y^{(k)}_{\ell,r}$ for all ℓ

    Solve (LP_k) with this r^k, let $y^{(k+1)}$ be the y-part;

    $N \leftarrow N + 1$;

  if $f(y^{(k)}) < f^\star$ **then**

    $f^\star \leftarrow f(y^{(k)})$,;

    $y^\star \leftarrow y^{(k)}$

for $t = 1, \ldots, T$ **do** `// Phase B`

  $r^\star \leftarrow$ argmax-pattern of $y^\star$;

  Pick $\kappa \in \{1, 2, 3\}$ cyclically; flip κ random rows of $r^\star$ to obtain $\tilde{r}$;

  Solve (LP_k) with $r^k = \tilde{r}$; let $y^{(0)}$ be the y-part; $N \leftarrow N + 1$;

  while *not converged* **do**

    Update r^k, solve (LP_k), update $y^{(k)}$; $N \leftarrow N + 1$;

  if $f(y^{(k)}) < f^\star$ **then**

    $f^\star \leftarrow f(y^{(k)})$,;

    $y^\star \leftarrow y^{(k)}$

return $(\tilde{y}, N) \leftarrow (y^\star, N)$;

Results

100 instances with $n = 10$, $m = 20$, and $v_{\ell i}$ drawn iid from $U[0, 10]$. Algorithm 1 is run with $S = 8$ random starts and $T = 20$

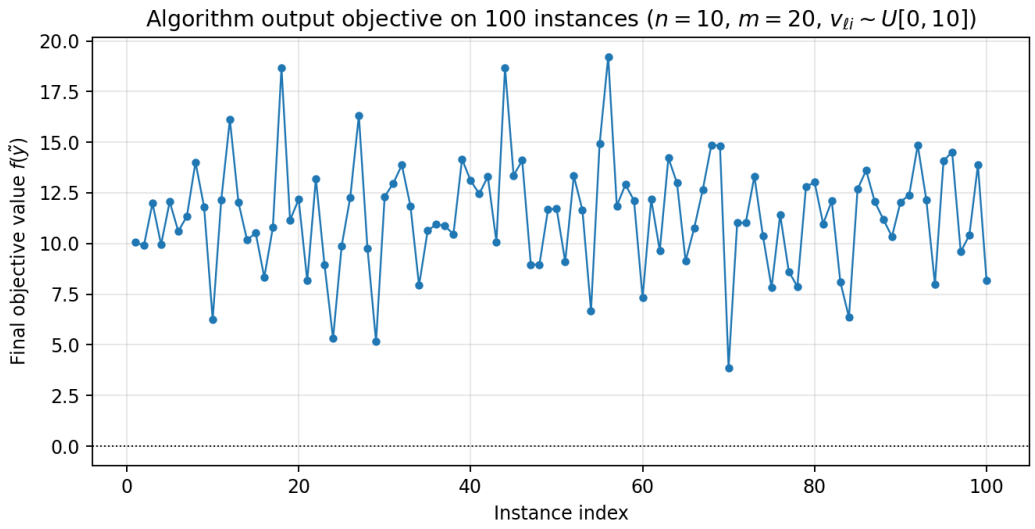


Figure 1: Final objective value $f(\tilde{y})$ returned by Algorithm 1 on 100 random instances with $n = 10$, $m = 20$ and $v_{\ell i} \sim U[0, 10]$ iid.

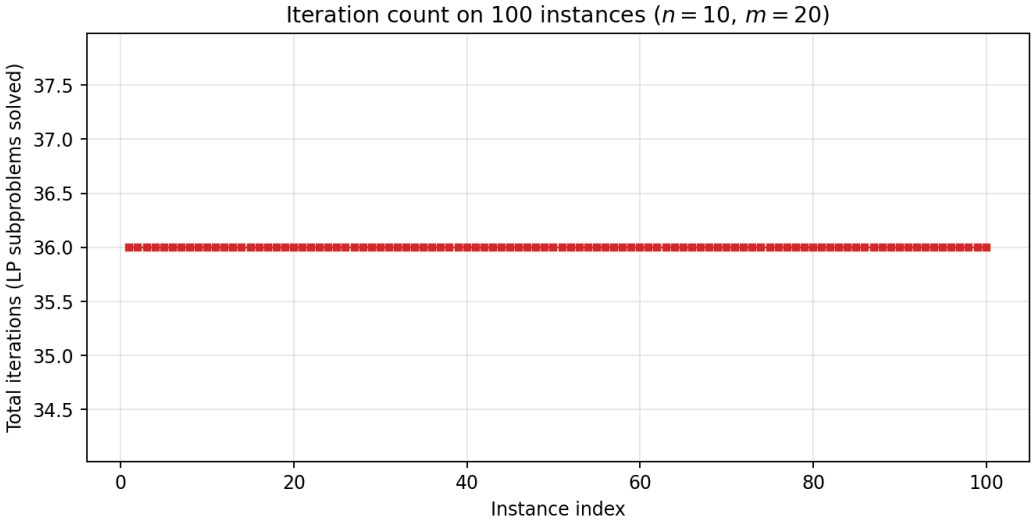


Figure 2: Total number of LP subproblems (LP_k) solved by Algorithm 1 per instance.

The returned objective values are concentrated between 3 and 18.